Chetna Kurzekar CS23174

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.preprocessing import StandardScaler
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
path="/content/drive/MyDrive/ML datasets/Mall_Customers (1).csv"
df=pd.read_csv(path)
df.head()
```

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Next steps: Generate code with df New interactive sheet

```
X=df.iloc[:,[3,4]].values
print(X)
```

```
[101  68]
```

```
 [103  17]
 [103  85]
 [103  23]
 [103  69]
 [113   8]
 [113  91]
 [120  16]
 [120  79]
 [126  28]
 [126  74]
 [137  18]
 [137  83]]
```

```
print(X_scaled)
```

```
[[-1.73899919 -0.43480148]
 [-1.73899919  1.19570407]
 [-1.70082976 -1.71591298]
 [-1.70082976  1.04041783]
 [-1.66266033 -0.39597992]
 [-1.66266033  1.00159627]
 [-1.62449091 -1.71591298]
 [-1.62449091  1.70038436]
 [-1.58632148 -1.83237767]
 [-1.58632148  0.84631002]
 [-1.58632148 -1.4053405 ]
 [-1.58632148  1.89449216]
 [-1.54815205 -1.36651894]
 [-1.54815205  1.04041783]
 [-1.54815205 -1.44416206]
 [-1.54815205  1.11806095]
 [-1.50998262 -0.59008772]
 [-1.50998262  0.61338066]
 [-1.43364376 -0.82301709]
 [-1.43364376  1.8556706 ]
 [-1.39547433 -0.59008772]
 [-1.39547433  0.88513158]
 [-1.3573049  -1.75473454]
 [-1.3573049   0.88513158]
 [-1.24279661 -1.4053405 ]
 [-1.24279661  1.23452563]
 [-1.24279661 -0.7065524 ]
 [-1.24279661  0.41927286]
 [-1.20462718 -0.74537397]
 [-1.20462718  1.42863343]
 [-1.16645776 -1.7935561 ]
 [-1.16645776  0.88513158]
 [-1.05194947 -1.7935561 ]
 [-1.05194947  1.62274124]
 [-1.05194947 -1.4053405 ]
 [-1.05194947  1.19570407]
 [-1.01378004 -1.28887582]
 [-1.01378004  0.88513158]
 [-0.89927175 -0.93948177]
 [-0.89927175  0.96277471]
 [-0.86110232 -0.59008772]
 [-0.86110232  1.62274124]
 [-0.82293289 -0.55126616]
 [-0.82293289  0.41927286]
 [-0.82293289 -0.86183865]
 [-0.82293289  0.5745591 ]
 [-0.78476346  0.18634349]
 [-0.78476346 -0.12422899]
 [-0.78476346 -0.3183368 ]
 [-0.78476346 -0.3183368 ]
 [-0.70842461  0.06987881]
 [-0.70842461  0.38045129]
 [-0.67025518  0.14752193]
 [-0.67025518  0.38045129]
 [-0.67025518 -0.20187212]
 [-0.67025518 -0.35715836]
 [-0.63208575 -0.00776431]
 [-0.63208575 -0.16305055]
```

```
scaler=StandardScaler()
X_scaled=scaler.fit_transform(X)
```

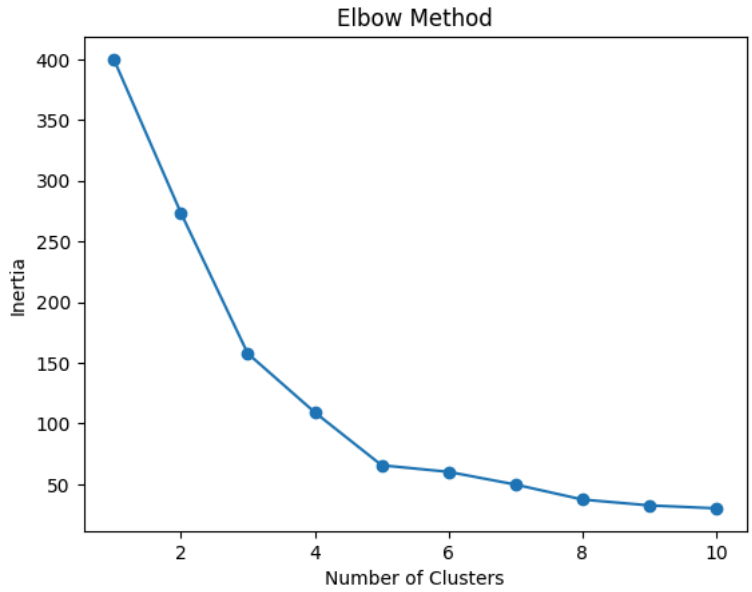
```
inertia_values = []
for k in range(1,11):
      kmeans = KMeans(n_clusters=k, random_state=42)
      kmeans.fit(X_scaled)
      inertia_values.append(kmeans.inertia_)

      print(inertia_values)
      plt.figure()
```

```
[399.99999999999994]
[399.99999999999994, 273.66888662642003]
[399.99999999999994, 273.66888662642003, 157.70400815035939]
[399.99999999999994, 273.66888662642003, 157.70400815035939, 109.22822707921345]
[399.99999999999994, 273.66888662642003, 157.70400815035939, 109.22822707921345, 65.56840815571681]
[399.99999999999994, 273.66888662642003, 157.70400815035939, 109.22822707921345, 65.56840815571681, 60.132874871934206]
[399.99999999999994, 273.66888662642003, 157.70400815035939, 109.22822707921345, 65.56840815571681, 60.132874871934206, 49.6
[399.99999999999994, 273.66888662642003, 157.70400815035939, 109.22822707921345, 65.56840815571681, 60.132874871934206, 49.6
[399.99999999999994, 273.66888662642003, 157.70400815035939, 109.22822707921345, 65.56840815571681, 60.132874871934206, 49.6
[399.99999999999994, 273.66888662642003, 157.70400815035939, 109.22822707921345, 65.56840815571681, 60.132874871934206, 49.6
<Figure size 640x480 with 0 Axes>
<Figure size 640x480 with 0 Axes>
<Figure size 640x480 with 0 Axes>
<Figure size 640x480 with 0 Axes>
<Figure size 640x480 with 0 Axes>
<Figure size 640x480 with 0 Axes>
<Figure size 640x480 with 0 Axes>
<Figure size 640x480 with 0 Axes>
<Figure size 640x480 with 0 Axes>
<Figure size 640x480 with 0 Axes>
```

```
print(inertia_values)
plt.figure()
plt.plot(range(1,11),inertia_values,marker='o')
plt.xlabel('Number of Clusters')
plt.ylabel('Inertia')
plt.title('Elbow Method')
plt.show()
```

```
[399.99999999999994, 273.66888662642003, 157.70400815035939, 109.22822707921345, 65.56840815571681, 60.132874871934206, 49.6
```




```
kmeans = KMeans(n_clusters=5, random_state=42)
cluster=kmeans.fit_predict(X_scaled)
df['cluster']=cluster
print(df.head())
```

```
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)  \
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40

   cluster
0        4
1        2
2        4
3        2
4        4
```

```
df.head()
```

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)	cluster
0	1	Male	19	15	39	4
1	2	Male	21	15	81	2
2	3	Female	20	16	6	4
3	4	Female	23	16	77	2
4	5	Female	31	17	40	4

Next steps: Generate code with df New interactive sheet

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 6 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
 5   cluster                 200 non-null    int32
dtypes: int32(1), int64(4), object(1)
memory usage: 8.7+ KB
```

```
df.shape
```

```
(200, 6)
```

```
print("Inertia:",kmeans.inertia_)
sil_score=silhouette_score(X_scaled,cluster)
print("Silhouette Score:",sil_score)
```

```
Inertia: 65.56840815571681
Silhouette Score: 0.5546571631111091
```

```
plt.figure()
sns.scatterplot(x=df['Annual Income (k$)'],y='Spending Score (1-100)', hue='cluster', data=df, palette='Set1')
centroids=scaler.inverse_transform(kmeans.cluster_centers_)
plt.scatter(centroids[:,0], centroids[:,1], marker='X', color='red', s=200, label='centroids')
plt.title('Customer segmentation using KMeans')
plt.legend()
plt.show()
```

```
kmeans = KMeans(n_clusters=4, random_state=42)
cluster=kmeans.fit_predict(X_scaled)
df['cluster']=cluster
print(df.head())
```

```
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)  \
0           1    Male   19                  15                      39
```

```
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40

   cluster
0        0
1        2
2        0
3        2
4        0
```

```
print("Inertia:",kmeans.inertia_)
sil_score=silhouette_score(X_scaled,cluster)
print("Silhouette Score:",sil_score)
```

```
Inertia: 109.22822707921345
Silhouette Score: 0.49434988482196784
```

```
kmeans = KMeans(n_clusters=7, random_state=42)
cluster=kmeans.fit_predict(X_scaled)
df['cluster']=cluster
print(df.head())
```

```
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)  \
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40

   cluster
0        5
1        2
2        4
3        2
4        5
```

```
print("Inertia:",kmeans.inertia_)
sil_score=silhouette_score(X_scaled,cluster)
print("Silhouette Score:",sil_score)
```

```
Inertia: 49.668244837367965
Silhouette Score: 0.50200146805547
```

```
kmeans = KMeans(n_clusters=8, random_state=42)
cluster=kmeans.fit_predict(X_scaled)
df['cluster']=cluster
print(df.head())
```

```
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)  \
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40

   cluster
0        4
1        3
2        4
3        3
4        4
```

```
print("Inertia:",kmeans.inertia_)
sil_score=silhouette_score(X_scaled,cluster)
print("Silhouette Score:",sil_score)
```

```
Inertia: 37.31912287833882
Silhouette Score: 0.4550112502601921
```

```
kmeans = KMeans(n_clusters=3, random_state=42)
cluster=kmeans.fit_predict(X_scaled)
df['cluster']=cluster
print(df.head())
```

```
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)  \
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40

   cluster
```

```
0        2
1        2
2        2
3        2
4        2
```

```
print("Inertia:",kmeans.inertia_)
sil_score=silhouette_score(X_scaled,cluster)
print("Silhouette Score:",sil_score)
```

```
Inertia: 157.70400815035939
Silhouette Score: 0.46658474419000145
```

```
kmeans = KMeans(n_clusters=3, random_state=42)
cluster=kmeans.fit_predict(X_scaled)
df['cluster']=cluster
print(df.head())
```

```
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)  \
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40

   cluster
0        2
1        2
2        2
3        2
4        2
```

```
print("Inertia:",kmeans.inertia_)
sil_score=silhouette_score(X_scaled,cluster)
print("Silhouette Score:",sil_score)
```

```
Inertia: 157.70400815035939
Silhouette Score: 0.46658474419000145
```